

in India for a minimum of 05 years from the date of submission of bid (not as joint venture, partnership firms or through any other association). In case of product failure OEM should replace malfunction product with equivalent working product immediately till the repaired or alternate product received.

- ix. IP CCTV System OEM for Cameras & VMS must be a member &/or listed in the ONVIF website. The quoted products must be ONVIF compliant not conformant. Online verification of OEM in ONVIF website must be available. No OEM should be banned or suspended by ONVIF within the last five years from the date of publishing the bid.

b. OEM of ICCC

- i. OEM of ICCC should have Installation base of at least 05 Safe City Projects / Smart Cities in India or globally during the last 05 years
- ii. OEM of ICCC should have supplied at least 3 + Sub System Integration in single order during the last 05 years in India or globally.
- iii. ICCC should have cyber-Security certifications from UL or equivalent from any Indian certifying laboratory (Suggestion: ISO 27001)

c. OEM of VMS

- i. OEM of VMS should have supplied at least 10,000 cameras Licenses in India or globally in qualifying orders during the last 05 years
- ii. OEM certifications
 - VMS Should be ONVIF Profile S & G.
 - Should also roadmap of ONVIF Profile T/Q. Declaration for the same to be provided.

d. AI Based Video Analytics/FRS/ANPR/Video Summarization Platform

- i. OEM must have supplied AI based Video Analytics platform in minimum 02 Safe Cities/Smart Cities/Surveillance project at Defense, Transport Segment, with a minimum 100 Cameras in a single city, single project with AI based Video Analytics use- cases projects in India or globally from last 05 Years.

e. OEM of Server

- i. OEM must have supplied servers in minimum 05 Safe Cities/Smart Cities projects in India
- ii. OEM of server should be validated with offered ICCC and VMS software with minimum one year of existence

f. OEM of Storage

- i. OEM should have successfully completed at least two order for supply and installation of minimum 10 PiB scale out NAS Storage in Safe Cities/Smart Cities projects in India.

g. OEM of Switches

- i. OEM of Switches should have supplied at least 1000 field switches in India into any Smart City/Safe City projects during the last 05 years
- ii. OEM should have authorized service center in India
- iii. OEM should have its own Repair/Service-Support center in the country and must have RMA set up in India for a minimum of 05 years from the date of submission of bid (not as joint venture, partnership firms or through any other association). In case of product failure OEM should replace malfunction product with equivalent working product immediately till the repaired or alternate product received.

- The system shall allow the users to perform advanced spatial analysis like geocoding, routing, buffering and attribute based analysis.
 - The application shall have standard and modern map navigation Tools
 - The system shall have the facility wherein the user can opt to view in 2D or 3D environment.
 - The system shall be compatible with Google Maps, Bing™ Maps, ESRI File Geodatabase, Micro Station, AutoCADG/Technology, ODBC source.
 - The System shall support hierarchical legends, and watermarks.
 - The application shall allow users to view the data with different symbology styles like differentiating feature records based on attributes or types, dynamic label generation with conflict detection, and translucency of all raster data and area color fill.
 - The system shall allow the user to find Address/Location
 - The system shall be able to consume real-time enterprise published spatial data. It shall be able to consume the third-party published OGC web-services.
- viii. Application shall be OGC compliant for database and shall provision conversion to other database formats.
- ix. GIS base maps shall be installed on work stations at Command control Centre and City Command and Control Center. GIS maps and data replication shall happen from central system remotely.

3.10.1. Video Analytics Usecases in the City

AI based Video Analytics usecases to be implemented on the surveillance CCTV cameras installed in the city. The advanced analytics system should be supported by continuous learning abilities. SI must implement the AI based video analytics on the GPU based servers in centralized architecture at datacenter.

The SI must provide AI based Video Analytics usecases as defined in the technical specifications to meet the requirements of the RFP focussing on the outcome, future scalability, security, reliability and adherence to SLAs and best practices in the industry.

4. Other Expectation and Consideration from MSI

4.1. Expectations from MSI/SI

- a) SI shall engage early in active consultations with the Authority, City Police and other key stakeholders to establish a clear and comprehensive project plan and understanding in line with the priorities of all project stakeholders and the project objectives.
- b) Study the existing fiber duct (if any) layout in the city and existing network to understand the existing technology adopted in each of the following areas (not limited to):
 - i. Surveillance Infrastructure – CCTV Cameras, Data communication, monitoring, control room and Infrastructure
 - ii. Other Smart City initiatives envisaged
- c) SI shall assess existing infrastructure's current ability to support the entire solution and integrate the same with the proposed solution wherever applicable and possible.
- d) SI shall judiciously evaluate the resources and time planned for undertaking the current state assessment, given the overall timelines and milestones of the project.

Systems Management Software	The server should come with systems management software to provide update management, configuration management, patch management and virtualization management.
Benchmarks	Server family should have published benchmark (Spec_int _rate2017)
Accessories	All the necessary tools & tackles licenses, cables/ connectors for Ethernet/ Fibre/ USB/ Power etc. required for making the system operational shall be provided by the bidder.
Industrial Standard Compliance	ACPI 2.0 Compliant, PCI 2.0 or higher Compliant, WOL Support, MS Logo Certification, USB 2.0 Support.

5.2.1.10. Functional & Technical Requirements for GPU Based Rack Servers (Video Analytics & FRS Servers)

Parameter	Specifications
Chipset	Latest Intel Chipset
Form Factor	Min 1U rack mounted with sliding rails
Processor	Dual Intel® Xeon® Scalable 2 nd Gen 16 Core 2.9 GHz processors
Memory slots	24 DDR4 DIMM slots, speed up to 2933MT/s, scalable to 3TB
Memory configured	Minimum 4 Gb per physical core
Disks supported	Drive bays: Up to 6 x 2.5” SAS/SATA (HDD/SSD)
RAID Controller	12Gbps PCIe 3.0 with RAID 1, 5, 10, 50 with 2GB Cache Memory
Disks configured	2x480GB SSD
DVD writer	DVD+RW
I/O slots	3 x Gen3 slots all x16
Network ports	2x 1G Ethernet and 2x 10G SFP+
Certified for OS	Windows Server 2016, VMWare, Red Hat Enterprise Linux, SUSE Linux Enterprise Server
Power Supply	Redundant Power Supply or higher
GPU Configured	3 x full height GPU (NVIDIA T4 or latest series) 8GB to be configured on Day 1

Management integration	Support for integration with Microsoft System Center, VMware vCenter, BMC Software
Power & temperature	Real-time power meter, graphing, thresholds, alerts & capping with historical power counters. Temperature monitoring & graphing
Pre-failure alert	Should provide predictive failure monitoring & proactive alerts of actual or impending component failure, memory, HDD
Configuration Management &	• Real-time out-of-band hardware performance monitoring & alerting • Out-of-band hardware & firmware inventory • Zero-touch auto configuration to auto deploy a baseline server configuration profile
	• Automated hardware configuration and Operating System deployment to multiple servers • Zero-touch repository manager and self-updating firmware system
LCD panel/LED Panel	As per OEM Design
HTML5 support	HTML5 support for virtual console & virtual media without using Java or ActiveX plugins
Server security	As per OEM Design
	Should provide effective protection, reliable detection & rapid recovery using: - Signed firmware updates - Secure default passwords
Warranty	5 years On-site comprehensive warranty with 24x7x365 remote hardware support.

5.2.1.11. Functional & Technical Requirements for AAA Server

S. No.	Parameter	Specifications
1	Servers	Should support approach that combines AAA, NAC, BYOD and Guest Access by incorporating identity, health, physical/device information, and conditional elements into one set of policies.
2		Must have ability to scale to up to 5000 devices per appliance from day 1
3		Solution must be Agnostic to existing wired, wireless and VPN network in place today.

57		Post login session statistics page displayed to users so they can monitor usage or quota assigned.
58		Support URL persistence so users originally requested webpage can be displayed post login.
59		Location based captive portal – display different landing page based on where guest is connecting to the network.
60		Support guest access across multi-vendor access networks.
61		Fully customizable self-registration or guest creation pages with user interface controls such as drop down, check list, radio button.
62		Authenticated self-registration for partner / joint venture account provisioning.
63		Published API's to allow 3 rd party system to manage guest accounts.

5.2.1.12. Functional & Technical Requirements for Continuous Learning Server
A.I/Training Server

Parameter	Specifications
Rack Height	4U
CPU Support	Must support 2 CPU's
Chipset	Intel C620 or better
Processors	Dual Intel® Xeon® Scalable 2nd Gen 16 Core 2.9 GHz processors
Memory	4x 32 GB RAM 2666 MT/s support up to 1500GB RAM, should have min. 16 DIMM slots
Hard Drives	4x3.84Tb SSD Should support up to eight hard disk drives (SAS, SATA, nearline SAS SSD: SAS, SATA)
GPU Configured	Should be configured with 8nos. Of Nvidia V100 / A100 32Gb GPU
GPU Support	Should support upto 10 GPU's
RAID Card	RAID Controller Card supports RAID 1, 5, 10
PCI Slots (I/O)	10 x PCIe 2.0/3.0 slots
NIC ports	2x1G & 2x10G baseT ports

Redundant Power Supply	Dual, Hot-plug, Redundant Power Supply
Management integration	Support for integration with Microsoft System Center, VMware vCenter, BMC Software
Power & temperature	Real-time power meter, thresholds, alerts & capping with historical power counters. Temperature monitoring
Pre-failure alert	Should provide predictive failure monitoring & proactive alerts of actual or impending component failure for fan, power supply, memory, HDD
Configuration & management	• Real-time out-of-band hardware performance monitoring & alerting
Management (continued)	• Automated hardware configuration and Operating System deployment to multiple servers
HTML5 support	HTML5 support for virtual console & virtual media without using Java or ActiveX plugins
Server security	As per OEM Design
OS	Windows server 2019 Standard Edition
Warranty	5 years On-site comprehensive warranty with 24x7x365 remote hardware support.

5.2.1.13. Functional & Technical Requirements for Blade Chassis

Feature	Specification
Chassis	Rack Mountable Chassis to accommodate Support for minimum 8 blade servers
Management Modules	Should support Hot Pluggable & fully Redundant Management Modules. The blade chassis should be configured with Hot swap IP based KVM Switch for Management or KVM Management should be integrated in Remote Management Controller.
Mid-plane	Should have passive mid-plane/no mid plane/ back-plane architecture
IO Connections	Hot swap and redundant cooling fans and all fans should be fully populated Dual end-to-end redundant Network connectivity for each blade

	• Removing an IP device • Enabling/disabling an IP channel • Refreshing an IP device (in case of firmware upgrade) • Multicast at multiple aggregation points
5	Community Surveillance Module:
.1	Architecture overview: a. The feeds city wide Surveillance system and feeds from Community surveillance (Invested by Public property owners in the premises) shall also be viewed at the Command and Control Center through the UC&C platform. b. It is envisaged that about 2500 cameras from community surveillance feeds would be extended from various Police Area Offices, Railway Offices, SP-Offices police stations across the city of Patna. c. The feeds from Community surveillance cameras could be fed into city UC & C platform through below 3 means and the solution must be ready to consume all the format of feeds and provide native intelligence within the UC & C platform – i. Through LAN / WAN from each community operator or a set of community operators to the nearest police station of each Police Area Offices, Railway Offices, SP-Offices and further federated to the Command & Control UC & C platform. The solution must be capable of viewing the streams both at the local police station levels or Viewing centers or other Police Area Offices, Railway Offices, SP-Offices police stations and Command centers. ii. Community surveillance camera connected via Public cloud network and further connected to the Command center UC & C platform.
5.2	The UC & C and VMS module must be flexible to adapt all the possible architectures outlined above and the operators must get a unified view of the CCTV feeds irrespective of the architecture through which the community camera feeds are extended to Police network.

7.4. Video Analytics Software for 1000 Cameras with average 2 use cases per camera

The followings are the AI based Video Analytis (nnot limited to) considering a smart City Scenarios

Violence & Violent Behaviour Activity:

- The human fighting.
- The human firing a weapon.
- The human throwing the stone.

- Person Tracking.

Women Safety Behaviour Activity:

- The women / person in distress raising his/her hand(s) for HELP.
- Chain / Mobile / Purse snatching
- The human lies or falls on the ground
- The human abandons an object.

Grouping Behaviour Activity:

- The Human or Humans group running.
- Humans Gathering in a group.
- Violation of Section 144 as per IPC

Vehicle Behaviour Activity:

- Vehicle Collision (such as Car, Bus, Truck, Motorcycle etc.)
- Vehicle Park on sidewalks or at no-parking areas. (Illegal Parking)

Other Activities

- Any kind of fire or smoke detection on the streets.
- Person Collapsing
- Crowd Estimation and management
- Camera Health Monitoring
- Abandoned Object Detection
- Advanced Intrusion Detection

7.5. Functional & Technical Requirements for Facial Recognition System

Sr. No	Key	ITEM DESCRIPTION
	Detection	Face Recognition System shall work on real time and offline mode for identifying or verifying a person from various kinds of inputs from digital image file and live video source from any IP video streaming sensor like IP Camera, Body Worn Cameras, Mobile handset cameras.
	Deep Learning Technology	FRS must be a latest generation Convolutional Neural Networks based facial and person tracking technology with Real-time 1:1 (one to one), 1: N (one to many) and N:N (many to many) matching.
	Live and Offline Mode	FRS shall be able to capture face images from live & pre-recorded CCTV feeds received and generate alerts if a blacklist (face from suspect list) match is found.

	Detections in crowd	The system shall be able work to detect more than 20 faces in crowd on moderate face rotation either horizontal or vertical. It should support a yaw angle of -40 to +40 degrees , a pitch angle of -30 to +30 degrees and a roll angle of -30 to +30 degrees.
	Detection of partial faces	The FRS shall recognize partial faces with varying angles from multiple videos simultaneously from Video clips, Group Photographs and VMS Playback directly from FRS Client Interface.
	Ability to add reference Images	The system shall be able to add photographs obtained from law enforcement agencies to the criminals' repositories tagged for sex, age, etc. for future searches.
	Support for cameras/video formats	The system shall support diverse graphic & video formats as well as live cameras. FRS shall support day/night operation with ability to detect faces both in colour and in black/white mode by using any H.264, H.265 Fixed IP Cameras with / without IR Illuminators. Bidder to ensure the identified cameras for FRS purpose are mounted and configured as per the splution design in order to achieve the desired results and accuracy
	User-management	FRS must support a user management module that enables different user level groups to support various permission levels.
		FRS client shall have ability to share recognition data like images & videos with multiple users and operators for better reference, alarm & incident management.
	Image Enhancement Capabilities	FRS system must have capability to enroll whatever images fed in the system with image enhancement and ability to verify the quality of the enrolled images with different colour indicator for low quality images enrolled in watchlist/database.
	Image Format support	The system shall be able to utilize any of the file formats like JPEG, PNG, BMP, TIFF etc. format for enrolment.

	De-duplication	FRS shall be able to check if new enrolled face is already enrolled in the database before registering the new enrolled face in the system. Also, the system shall be able to find a previous detection of a POI (person of interest) upon enrolment to watchlist (retrospective search) in less than 5 sec.
	Enrolment of faces	The system shall have option to automatically enroll face images from CCTV cameras/video source. The system should also have an option for Bulk Enrollment either from file system or a 3rd party databases such as UID, SAARTHI, IT, NCRB, EPIC etc. The end user shall provide required interface with such Govt databases for integration purpose to the qualified bidder.
	Categories of database faces	The system shall have capacity to create different categories of people with option to customize the matching threshold for different categories.
	Full HD Support	The system shall be able to work on full HD Camera video with maximum performance.
	Implementation	The system shall be able to be implemented on IT hardware like Server or Workstation.
	OS Support	The FRS algorithm should be able to use proven open source tools and technologies like Linux to bring down the total cost of ownership of the solution. FRS running on any other OS should be supplied with Pre-Licensed Server based latest version OS like Microsoft Server 2016 and Microsoft SQL as needed by the application
	Database Support	The system shall employ database system like MS SQL/ MYQL/ Leading Open Source Database/Sybase/ Mongo DB/ Postgres/Oracle etc. The FRS system should natively integrate with Video Intelligence platform and use a common database of the platform, so that common queries can be made on the common database for faces detection and other events.

	Algorithm Benchmarking	The Vendor should have any performance benchmarking certificate. NIST certificate will be preferred.
	Performance	The system must perform a full 1: N search of the probe image in under 5 seconds against a database of up to 1 mn face records.
	Mobile Application Support	FRS Software vendor shall have mobile application of the same FRS software to support iOS and android based smart field devices. Mobile application shall be capturing the face of suspect in field and sending back to the FRS server for matching. Matching result shall be shown on the mobile application screen with matching score. There shall be provision in mobile application to stream mobile device camera as video streamer.
	Detection robustness	System shall be able to detect the faces across the multiple CCTV video sources for online (real-time) and offline modes regardless of following conditions:
		a. Changes in Facial expression
		b. Changes in facial hair or hairstyle
		c. Changes by moderate aging (up to 15 years)
		d. Partially hidden faces or occluded faces like wearing dark glasses mask etc.
		e. Changes in lighting conditions
	Search Capabilities	Simple Search UI that facilitates quick and easy access to the collection of events recorded by the system without the constant monitoring by operators and must perform a full 1: N search of the probe image in under 5 seconds against a database of up to 5-8 Million POIs. It shall support following
		a. Search previous events by images from previous detections
		b. Search previous events by images uploaded by operator
		c. Search previous events by enrolled names
		d. Search previous events by date and time

		e. Search previous events by watchlist group
		f. Search in Watchlist by image
	Retrospective Search	FRS shall have capability of Search backwards for previous detections and/or recognitions (events) of the detected person without enrolment from live CCTV & other forensic videos / offline videos
	Upto 5 nearest matches support	FRS shall have ranking features to show next 5 closest & similar subjects in the Watchlist with nearest score to the detection. This option enables you to review POIs that are potential matches for this detection for efficient system performance.
	OEM owned algorithm	The FRS OEM should have ownership of Face Recognition Engine /Algorithm for any custom specific development as required by client
	Map feature	FRS must allow tracking of person on maps to be uploaded in the system for cameras connected to FRS and shall highlight the camera location on the map for each detection/alert.
	SDK/API for integration	FRS shall provide an SDK/API for integration with any third-party software like ICCC Command & Control Centre. API must be available with a full set of documentation of each method with accompanying sample code. All FRS function shall be fully accessible via API.
	Timeline of detections	FRS shall provide timeline sequence of all detections of subject with date, time & location.
	Email Integration	FRS shall support email Alerts via Gmail, Outlook or via an Exchange SMTP service. Different recipients can be defined for different Camera Groups. User shall be able to define how frequently recognition/detection emails are sent, the email subject and the email sender (among other things). The email itself includes the timestamp of the detection, the score, the description, the reference image (defined in the Watchlist) and the detected image.

Use of AI accelerator hardware	FRS shall use extensive AI Technology and perform video processing on GPUs like NVIDIA; INTEL or similar as per design & sizing vetted by AI FRS Algorithm OEM. The number of servers to be supplied, shall be based on the number of camera channels on which the FRS needs to be performed as per OEM design considerations.
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Face Recognition System KPI Criteria/Evaluation Parameters:

Sr. No.	Aspect
2	Measuring following KPI (Key Performance Indicators) Detection Rate Number of True Positives Number of False Positives Number of True Negatives Number of False Negatives All vendors will be provided with a live stream with similar Field of View to ensure common ground along with suspects who can be enrolled. Above outcomes will be measured and compared among all vendors to check the accuracy.
3	Simultaneous detection of multiple faces in crowd: All vendors will be provided with a crowded video and outcomes will be recorded.
4	Deep Learning based algorithm: All vendors will have to demonstrate learning capabilities within their algorithm.
5	Global Threshold, Camera wise threshold and watchlist wise threshold
6	Real Time back search for newly enrolled subjects
7	Video & Image evidence of suspects
8	Easy Monitoring of System Health
9	Multiple detections to be collated using intuitive methods
10	Privacy as per GDPR compliance
11	Integration with leading VMS vendors
12	Integration with any sensor

7.6. Video Summarization Functional & Technical Specification:

	Functional Requirement Picture Intelligence Unit (PIU) will maintain all Audit Trails and logs of all instances, triggers and incidental data derived out of VA platform, FRS Engine, ANPR Engine and the activities governed by the VMS. PIU will not replicate the incidental data however, will keep the track record of all such activities as a common directory which provide convenience during post investigation analyses. Picture Intelligence Unit (PIU) shall be given access to the databases of Passport, CCTNS, Prisons, AMBIS or open source databases or any similar database available with the State Government and Central Government. It shall also create a repository of photographs obtained from various police sources like Newspapers, Photos during raids, Photos sent by people, etc. Such photographs would be tagged for sex, age, etc. so that these become searchable. Some of the approaches to be used by the Picture Intelligence Unit for implementation of Facial Recognition System would be as follows: a) Match a suspect/criminal photograph with these databases. b) Search photographs from the database meeting certain suspect features. c) Match a suspected Criminal face with Video Feeds of specific camera locations or with the feed received from private or other public organization’s video feeds. d) Whenever there is a requirement at any of the Police team to check the identity of an individual, Police Station would make such a request to PIU, who in turn would search the databases available with it to match the individual. Such databases would either be accessed through web services or in a downloaded manner. Full audit trail of reports and data provided will be maintained. e) PIU would also try to access information from third parties — for example, banks, telecom companies, credit companies, etc. PIU shall also liaison with other institutes & agencies doing R&D work on Facial Recognition & related technologies. It is expected that the Systems Integrator look at this Picture Intelligence Unit as a Research & Development Centre to test the best and latest technologies and ensure its continuous enhancement. SI is required to analyze the futuristic requirements for the effective functioning of PIU and propose tools like Data mining (application of statistical techniques and programmatic algorithms to discover previously unnoticed relationships within the data). f) Using the Key UI and integration capabilities of the Video intelligence platform, PIU shall oversee the integration of ANPR with the other relevant databases
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7.8. AI with continuous Learning & Improvement System - Functional Requirement & Technical specification

S.no	Parameter	Specifications
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1	Advanced AI compatible	The system shall be an AI/Deep-learning based Behavioral understanding systems for Real-Time video intelligence in the most dynamic and challenging urban environment on a 24x7 basis.
2	Training new models	The system shall be futuristic and keep improvising its accuracy on the existing behavior of diverse objects, and allow it to add more behaviors in the upcoming versions.
3	Annotation	The system shall have an inbuilt annotation tool that allows a user to label the images with relevant information using both rectangle and polygon drawing facilities. The annotation should allow labeling of images or drawn objects with different class names. In case of persons, it should also support labeling of various attributes such as color of clothing, type of clothing, age, gender etc. as well. The user-interface should allow to plug-in the trained model in any of the relevant Video Analytics use-cases dynamically at each camera. The system should allow the user to plug newly trained AI models at runtime by simply selecting the models in the per-camera configuration page
4	Model Comparison	The System shall have a library of standardized AI models developed by the OEM of the Video Analytics System. These models shall be used for comparing and benchmarking the performance of newly developed models. The system shall allow for both qualitative and quantitative comparison of models, i.e. it shall allow the end user to compare individual parameters of the model (such as learning rate) as well as the overall performance of the model on any given dataset when compared to a standardized model.
5	Monitoring and analytics	Autonomously objective metrics shall be available to be evaluated and Insights into the performance of each algorithm, model and their versions shall be made available to key stakeholders or users as defined. Visual map of composition, workflow, usage analytics, resource utilization, failure points etc. would be made available to provide complete control of A.I. workload.
6	Supervised deep learning methods	The system shall be automated and does not require typical rule or filter drawings to detect any interest in alarm.
7	Self-learning Capabilities	The system shall apply holistic and temporal analysis for scene participants through a top-down approach (scene -> event(s) -> action(s) -> object(s) -> attribute(s) by using the time dimension to connect between the video's image sequence and the different modalities.

8	Terminology	The system shall detect objects, actions, events, and scenes where the size of the participating objects is at least 10% of the frame. The objects and behaviors should not be obscured.
9	Applying Multiple Activity Rules	The system shall allow applying multiple or all behaviors on a camera or a set of behaviors on the camera group using a simple user interface also shall detect any changed behaviors within 03 seconds in front of the camera once the datasets are fully trained.
10	Key UI View and functionalities	The System shall provide the following key results from the use case
		Event Notifications: The result of each of the use case shall be in the form of events that contain the screenshot with other metadata describing the event, such as detected objects, timestamp, camera/video that generated the event and all other metadata representing the event from different use cases. The User Interface shall have a grid and list view with all the events from different use cases, cameras etc. These features should be supported through a mobile application to be utilized by various field users. The system should support customization of alerts, video feeds, and priority-based alerts for individual users from day one.
		Resource Management View: The User interface shall provide a list of all the resources available in the system such as computing servers and cameras. The status of each of the devices, whether they are online/offline shall also be available at all times.
		AI Training Tool: The User interface shall have a training tool to annotate and label images from the events to train new AI models and update the existing ones. The training tools shall also contain a list of all the models available in the system, which can be plugged into any AI use case easily.
		Use case deployment matrix: The user interface shall have a matrix to assign, start, stop and schedule any use case on any camera. The status of active and non-active use cases shall be clearly visible with colour coded information. All the licenses should be able to operate in floating mode for all cameras.
		Data Analytics Dashboard: The user interface shall also have an analytics dashboard listing all the patterns of events from different cameras with a heat-map of number of events on an hourly basis.
		The user interface shall be a unified dashboard that shows events from all the Video Analytics use-cases and all the cameras in a common UI, and which gets populated in real time from event notifications.

		The User interface of the system shall be a web interface that can be accessed from any system in the local area network with login credentials. It shall allow multiple users to log in at the same time, and receive real-time alerts and notifications.
11	Common UI for all the use-cases	The system shall be seamlessly integrated with the proposed VMS and showcase the generated metadata on the live & playback video stream.
12	Web based Interface	The system shall allow each use-case to be uniquely configured for every individual camera stream, with parameters for camera calibration, image quality improvement, night/day settings etc.
13	Live Video Interface	Each use-case shall be able to run on different cameras with different settings (e.g., different Zones for Intrusion, different lines for line crossing detection, etc.) at different hours of the day.
14	Configuration per-use-case per-camera level	The configuration page shall allow a user to choose any of the available AI models to detect and classify objects within the image. The description of the models shall clearly specify performance and hardware requirements of each of the model.
		The system shall be ready to deploy and does not require any additional camera calibration.
		The use case on each camera shall allow setting up configuration of multiple detections zones such as lines and regions that can be used to define perimeters, regions of interest.
15	Camera Calibration Tool	The configuration user interface shall allow adjusting various sensitivity and confidence parameters to adjust each video-analytics use-case's performance with respect to the physical deployment of the camera.
16	Key configuration parameters	The system shall allow a user to filter and retrieve all the events based on any combination of the following parameters: - Time of the event - Objects in the event - Type of the use-case - Camera Location etc.
		The architecture shall clearly demonstrate the technology stack with layers of the core platform, data governance and interface to different software applications.
17	Filtering and Retrieval	The system must have a consolidated alert dashboard where each alarm type can be accompanied by a video snippet in a form of different color codes.

18	Transparent and Open Architecture	The system shall support user with a hierarchical access level, with different access level for different users demarcated with respect to cameras, locations and the data. The user access control system shall allow setting of SOP's like CRUD (Create, Read, Update and Delete) operations for each user.
19	Highly parallel and distributed	The system shall allow deployment of any use case on any camera without any MAC level or IP level locking. Ideally any use case shall be deployable and redeployable on any camera or video source as far as the camera view supports such use cases to be deployed.
20	User Management	The System shall be a real-time video analytics engine that utilizes advanced image processing algorithms to turn video into actionable intelligence. The AI based Video Analytics system shall consist of video-processing & analytics engine that works seamlessly both on saved videos or camera streams in real-time and provide events to the user based on the use-case basis. The system shall be compatible with all ONVIF compliant IP cameras with H.264/H.265 video decoding.
21	Deployment of use-case across any camera	All the video streams shall be processed centrally at the data centre with one or more servers for video processing. The user shall be able to log in to the system through the central dashboard to access all the data from all the servers. The processing of videos as well as alert generation shall be done on premise. At no point in time shall the data from the site be shared over the internet or sent over to the cloud. The System UI shall only be accessible using workstations and terminals available on premises.
22	Video Compatibility	The AI system shall also support third-party developed algorithms and use-cases that can provide the user with a large base of use-cases to choose from.
23	Centralized Deployment Support	If a new use-case needs to be developed based on Video Intelligence, the system shall provide a developer API / Software Development Kit (SDK) for this purpose. The API / SDK shall be provided along with detailed documentation for building end-to-end use-cases on the system.
24	Support for third-party use-cases	The system shall also allow the user to plug different AI models in the individual running of the video analytics use-case.
		The technology stack shall be modular and scalable based on containerized micro services. Each use-case shall be orchestrated as a stand-alone micro service, which communicates with a central server for exchanging of the data.
		A.I. micro services components shall be agnostic to language used in technology stack. It shall work with any language, framework, and library of choice without any impact on the rest of the architecture.

		This type of flexibility will ensure lower friction for collaboration and deployment of AI.
25	Flexible Technology Stack	Algorithms being containerized shall ensure both interoperability and portability, allowing for code to be written in any programming language or any version of library and framework but then seamlessly exposes a single API to be integrated and ported with multiple modules/AI components of diverse stack. It shall seamlessly integrate with other components and shall be portable/ replicable easily across the machines automatically.
		The Video Analytics shall be based upon Machine Learning and Deep Learning framework.
		To save the duplication of the video storage, the analytics should flag the video for the configurable duration of time pre and post event in the Video Management System. It should be possible for the operator to jump to the alert flag in the archived video for detailed investigation of the event.

7.9. Functional & Technical Requirements for Outdoor Fixed Cameras/Bullet/Dome(HD)

S.No	Features	Specifications
1.	Form Factor	Box Type / Bullet Camera
2.	Image Sensor	1/2.8" Progressive CMOS or better
3.	Day/NightvOperation	ICR with IR range of 100m or better
4.	Minimum Illumination	Color 0.10 lux , B/W 0.0005 lux
5.	Lens	Motorised Lens (5 mm to 50 mm / 12 mm to 40 mm or better) or as per field requirement to achieve the required FoV
6.	Electronic Shutter	1 ~ 1/10000 sec.
7.	Image Resolution	1920X1080 @ 30 fps (2MP)or better
8.	Compression	MJPEG, H.265,H.264 or better
9.	Frame Rate and Resolution	Full HD (2MP 1920x1080 or better) @ 25/30 FPS
10	Simultaneous Stream	Minimum 3 streams should be configurable at 1920 X 1080 @ 25 fps simultaneously
11	White Balance	Auto / Manual / ATW / One Push
12	Noise Reduction	3DNR / 2DNR / Color NR
13	Zoom	Digital Zoom
14	Video Streams	Three Stream supportable , All stream should be H.265
15	Image Setting	Saturation, Brightness, Contrast, Sharpness, Hue adjustable

ANNEXURE 3: ANALYTICS USE CASES REQUIRED WITH THE TYPE OF LOCATIONS

(Related Corrigendum)

Sl. No.	Type of Analytics and Location	As per RFP	Revised As	Video Analytics Use case
1	Vehicle Related	150	250	Vehicle Classification/ Vehicle Detection and Video Capture Module
				Vehicle Detection by Color
				Hot Listing and Alert Generation
				Use of Mobile phones by Driver while driving /Red Light Violation Detection
				Wrong side driving detection/ Vehicle Behaviour Activity: •Vehicle Collision (such as Car, Bus, Truck, Motorcycle etc.) •Vehicle Park on sidewalks or at no-parking areas. (Illegal Parking)
				No helmet violation detection
				Triple ride Violation Detection
				Speed Violation Detection /Vehicle Wrong Direction Violation Detection/ Vehicle Left Side Violation Detection
2	Video Analytics Use Cases	100	200	Violence & Violent Behaviour Activity: •The human fighting. •The human firing a weapon. •The human throwing the stone. •Person Tracking.
			200	Women Safety Behaviour Activity: •The women / person in distress raising his/her hand(s) for HELP. •Chain / Mobile / Purse snatching •The human lies or falls on the ground •The human abandons an object.
			100	Grouping Behaviour Activity: •The Human or Humans group running. •Humans Gathering in a group. •Violation of Section 144 as per IPC
3	Surveillance Related at Property of interest/Other Analytics (Bus Stop, Important Buildings, Monuments, Parks, Stadium, Tourist Locations, Education Institutes, etc.)	200	250	Overcrowding Detection/conflicts in crowd
				Abandoned Object Detection
				Vandalism Detection
				Intrusion Detection/ Camera Health Monitoring
				Person Collapsing
				Loitering
				People Counting
				People Tracking across cameras